XoYokeBridge

Logitech Flight Yoke & Rudder Bridge for Xbox Flight Simulation
Complete System Operator Manual

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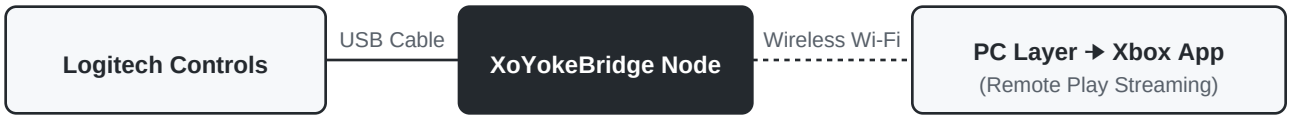
1. System Overview & Data Architecture

The XoYokeBridge system acts as a high-speed wireless translator between your enthusiast-grade flight hardware and the Xbox ecosystem. This setup eliminates long, messy USB cables cutting across your room, allowing you to completely hide your processing computer while maintaining an authentic, near-zero latency cockpit experience.

The system processes and routes your flight data through a distinct four-stage pipeline:
• **Hardware Capture:** The XoYokeBridge hardware unit connects directly to your physical cockpit controls (Logitech Yoke, Throttle, Rudder combo), instantly reading your mechanical movements, switches, and pedal inputs with precision.
• **Wireless Streaming:** Operating over your local home Wi-Fi network, the hardware automatically scans your environment to locate your local computer and beams the raw, real-time control data wirelessly across the room.
• **PC Virtualization Layer:** A background utility running on your PC receives this wireless stream. The software cleans, stabilizes, and calibrates the control signals, converting them instantly into a virtual local gaming controller format.
• **Xbox App Routing:** The PC passes this virtual controller data directly to the official Xbox App, which is running a Remote Play session with your Xbox console. The Xbox console reads the inputs as a native, fully responsive controller, allowing you to pilot your aircraft seamlessly inside Microsoft Flight Simulator (MSFS).



2. Hardware Configuration & Physical Setup

Connecting Your Controls

Your auxiliary flight controls must daisy-chain directly through the main yoke infrastructure rather than connecting to an outside computer:
1. Connect your Logitech Throttle Quadrant and Logitech Flight Rudder Pedals directly into the available expansion ports on the back of the Logitech Flight Yoke.
2. Take the primary, main USB communication cable coming out of the Logitech Flight Yoke and plug it directly into the dedicated input port on your XoYokeBridge hardware unit.

Critical Power Requirements

The Logitech Flight Yoke system demands a stable, dedicated power source to properly register all auxiliary switches, levers, and pedals. You must power the Logitech Yoke using an external 5V power supply plugged into a wall outlet. Avoid powering the infrastructure via unpowered USB hubs or unstable ports, as insufficient voltage will cause control dropouts or signal calibration errors during flight.

Physical Modification Note (3.5mm Jack)

If your setup utilizes the integrated 3.5mm accessory jack, please note that the thick plastic casing of the Xbox-style housing tightly constricts the entry port. Because of this structural design, standard 3.5mm cable audio jacks can be incredibly difficult to insert fully, often failing to click into place.
To achieve a secure physical connection, you will need to perform a minor modification. Carefully use a drill or a small hobby knife to widen the plastic opening on the device housing, or shave down the outer rubber/plastic sleeve molding of your 3.5mm jack cable until it fits cleanly inside the recess.

3. First-Time Setup & Network Provisioning

When the XoYokeBridge hardware is powered on for the very first time, or whenever it is unable to locate a previously saved home wireless connection, the system automatically defaults into Setup Mode. This automated configuration process securely binds the hardware to your home network without requiring specialized programming cables.
• **Broadcast State:** Upon entering Setup Mode, the front status LED will begin flashing in a rapid pattern, indicating that the device has launched its localized setup server. It will broadcast a temporary wireless network named FlightBridge_Setup.
• **Accessing the Portal:** Using a Wi-Fi-enabled device such as a smartphone, tablet, or laptop computer, open your wireless settings and connect directly to the temporary FlightBridge_Setup network. Once connected, follow the automated on-screen setup page that initializes in your device browser.
• **Network Selection:** The bridge will immediately perform a localized scan of nearby Wi-Fi networks and display them in a scannable on-screen list. Simply select your preferred home Wi-Fi network from the list, input your network password, and save the settings.
• **Automated Deployment:** Once a successful connection is verified, the XoYokeBridge hardware locks the credentials into its persistent internal storage, shuts down the temporary configuration portal, automatically reboots the internal modules, and returns to normal operation.
In most everyday scenarios, this configuration loop only needs to be completed once. The hardware safely stores this information indefinitely unless you swap out your home internet router, modify your primary Wi-Fi passwords, or intentionally trigger a full factory reset.

4. Core System Capabilities

• **Plug-and-Fly Automation:** Once configured to your home Wi-Fi network, daily operation requires no manual calibration or step-by-step pairing. Simply apply power to the bridge, boot up your flight stream, and take off.
• **Persistent Network Memory:** Your Wi-Fi network names, passwords, and previous successful connection paths are saved securely inside internal storage. The system retains this data indefinitely, even after complete power loss or extended storage.
• **Flight-Safe Signal Protection:** If your home Wi-Fi drops out or suffers from sudden interference mid-flight, the system instantly engages a localized safety buffer. Your flight controls are automatically brought to a safe, neutral level and the aircraft throttle is throttled back to idle. This completely prevents un-commanded dives or stalls, automatically restoring full piloting authority the millisecond the wireless network reconnects.

5. LED Status Reference

The front panel features a smart status indicator light suite to clearly communicate what the internal systems are doing without exposing technical software configurations. An Amber Light tracks hardware Power state, while the primary Blue LED captures status tracking rules:

• **Fast Flashing:** Initial setup state. Unit is hosting its configuration portal, waiting for credentials.
• **Slow Pulsing (1s Interval):** Active network discovery mode. Searching for your saved home Wi-Fi profile.
• **Ultra-Slow Pulsing (5s Interval):** Connection handshake idle. Linked to Wi-Fi and waiting for the PC software layer.
• **Solid On:** Active link operational environment. Control surfaces are streaming data natively to the simulator layout.
• **Double Flash Pattern:** Connection credential mismatch error. Your Wi-Fi name or password was rejected.

6. Troubleshooting & Maintenance

• **Status Light Stays in a Slow Pulse:** The hardware cannot find your router framework. Ensure your Wi-Fi loop is active and move the node into clearer signaling distance bounds.
• **Status Light Stays in an Ultra-Slow Pulse:** The hardware is online but isolated. Ensure your local host computer is turned on, bound to the same network ID layer, and that the companion virtualization software utility execution has been launched. For cleanest handshake routines, always boot up the hardware node before starting the software framework.
• **Flight Controls Randomly Stop Responding:** Signifies temporary environmental wireless dropping blocks. Keep your flight lane unchanged; your system's localized safety algorithms will recover and re-engage manual control frames within seconds.
• **Moving to a New Wireless Network:** Perform a localized Factory Reset operation to clear old configuration tables out of the unit's secure storage, returning it to day-one setup bounds.

XoYokeBridge PC Companion Software Manual

Automated Configuration & Link Utility for Xbox Flight Simulation

PC-1. System Overview

The XoYokeBridge PC Companion Software is a lightweight background application that completes the wireless link between your flight hardware and your flight simulation environment. While the XoYokeBridge hardware registers your hands-on inputs on the flight deck, the PC software automatically captures this stream over your home network, seamlessly matching your flight equipment to the Xbox ecosystem.
The software is engineered to be virtually invisible during your flight sessions. It runs quietly in the Windows System Tray with a minimalist footprint, automatically waking up when you start your simulator and shutting itself down cleanly when your flight session or Xbox Remote Play stream is closed.

PC-2. Core Features & Automatic Protections

• **Automatic Hardware Discovery:** The software handles all network configurations internally. When launched, it automatically scans your local home network, locates your active XoYokeBridge hardware, and establishes a secure connection. You never have to type or manage confusing network settings.
• **Smart Reconnection Handling:** If your home Wi-Fi network experiences a brief signal drop, or if your flight computer or hardware unit is unexpectedly restarted, the software automatically loops in the background to restore the flight stream without requiring user intervention.
• **Flight-Safe Guardrails:** To protect your aircraft from unpredictable behavior during local wireless interference, the software features an automated 3-second safety tracker. If communication from the flight deck drops for more than 3 seconds, your aircraft controls are automatically stabilized to a perfectly safe, level neutral position, and the engine power is brought back to idle. Full manual piloting control instantly returns the millisecond the connection stabilizes.
• **System Optimization Enforcer:** The application includes built-in protection checks that prevent duplicate instances from opening if you accidentally double-click the shortcut. It also manages its own diagnostic log sizes automatically, ensuring the background application never takes up unnecessary space on your computer's storage drive.

PC-3. Flight Deck Layout & Operational Mapping Blueprints

The software automatically syncs your premium flight hardware directly into standard Microsoft Flight Simulator controls. No manual configuration or button remapping is required.

Axis & Servo Mappings

Flight Hardware Input	Code Axis Key	Virtual Xbox Target
Yoke Roll	"Roll"	Left Stick X-Axis
Yoke Pitch	"Pitch"	Left Stick Y-Axis
Yaw (Rudder)	"Yaw"	Triggers (RT / LT)
Throttle Lever	"Throttle"	Buttons A / B
Brakes	"Left/Right_Brake"	Button X
Prop / Mixture	"Propeller" / "Mixture"	None

Digital Button & Switch Mappings

Flight Hardware Input	Code Key	Virtual Xbox Target
Left Red Button	"L_Red_Btn"	X Button
Right Red Button	"R_Red_Btn"	Right Thumb Click
Left Rocker Switch	"L_Rocker_UP / DN"	Unassigned
Right Vertical Switch	"R_Vert_UP / DN"	D-Pad UP / DOWN
Right Horiz. Switch	"R_Horiz_LF / RT"	D-Pad LEFT / RIGHT
Toggle Switch 1 (UP)	"T1_UP"	A Button
Toggle Switch 2 (DN)	"T2_DN"	B Button
Toggle Switch 3 (UP)	"T3_UP"	Left Thumb Click
Toggle Switch 4 (DN)	"T4_DN"	BACK Button
Toggle Switch 5 (UP)	"T5_UP"	START Button
Hat Switch (POV)	"POV"	Right Stick 8-Way

Inactive Hardware Passthroughs

Flight Hardware Input	Code Key	Virtual Xbox Target
Toggle Switch 6 (DN)	"T6_DN"	None
Detent Throttle Lever	"Detent_Throt"	None
Detent Propeller Lever	"Detent_Prop"	None
Detent Mixture Lever	"Detent_Mix"	None
Mode Dial Switch	"Mode"	None

PC-4. Installation & Basic Setup

System Dependencies & Prerequisites

To create a high-fidelity bridge between premium flight systems and your Windows streaming environment, the XoYokeBridge deployment framework relies on a core system-level driver. To make your setup as effortless as possible, the installer automatically handles this requirement:

• **ViGEmBus Framework:** An industry-standard virtual controller emulation driver that registers flight hardware inputs directly into the Windows kernel system.

Note: This component carries an official, verified Microsoft digital signature, is trusted globally by the simulation hobbyist community, and requires zero secondary configuration from the pilot. Your core application handles everything seamlessly out of the box.

Installation Steps

1. Run the primary support driver installer on your Windows machine to initialize the ViGEmBus framework.
2. Launch the main XoYokeBridge software setup configuration tool wizard to install the application.
3. If Windows Defender or your local Firewall prompts you upon first launch, select **Allow Access** on your private home network. This gives the application permission to safely scan your local network for your hardware unit.

Recommended Startup Order

To ensure the fastest automated handshake across your entire wireless flight layout, adopt this simple pre-flight routine:

1. Turn on your XoYokeBridge hardware unit and wait for its indicator light to signal that it is ready.
2. Launch the XoYokeBridge companion software application on your Windows PC.
3. Open your Xbox App Remote Play connection and start Microsoft Flight Simulator.
4. Verify your indicators show an active connection, complete your standard pre-flight checklist, and enjoy your flight.

PC-5. Troubleshooting & Diagnostic Interface

• **Controls Do Not Respond:** Ensure your flight simulation software or Xbox Remote Play stream is fully running on your screen. The Xbox app quits sending data to the console as soon as it goes out of focus or is minimized. If the game is open but controls are frozen, check that your initial framework installation completed successfully, and restart your PC to ensure all system updates have taken full effect.

• **Interactive Verification Panel:** If you ever want to verify your equipment calibration or check your network range, you can open the built-in Diagnostics Terminal Display Panel. This safe interface lets you visually check your system's health without touching any code: Real-time network streaming refresh updates (measured in Hz), active visual movement gauges for your Roll, Pitch, and Rudder axes, and live tracking indicators that light up when you click your physical buttons or flip switches. This window is built purely for physical calibration testing and troubleshooting; you can safely leave it closed while flying.